

2021-2022 HS Virtual Challenge Meet #1 - Science

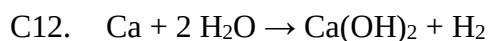
ANSWER KEY

B01. E	C01. B	P01. C
B02. D	C02. D	P02. E
B03. A	C03. C	P03. A
B04. B	C04. A	P04. D
B05. B	C05. B	P05. B
B06. A	C06. E	P06. C
B07. D	C07. A	P07. D
B08. D	C08. E	P08. E
B09. D	C09. C	P09. E
B10. B	C10. C	P10. B
B11. D	C11. C	P11. C
B12. A	C12. B	P12. A
B13. C	C13. E	P13. D
B14. D	C14. E	P14. B
B15. A	C15. E	P15. E
B16. A	C16. A	P16. C
B17. D	C17. D	P17. C
B18. B	C18. A	P18. A
B19. A	C19. B	P19. D
B20. C	C20. E	P20. A

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SELECTED SOLUTIONS

C06. $270g\ C_6H_{12}O_6 \left(\frac{1\ mol\ glucose}{180\ g\ glucose} \right) \left(\frac{6\ moles\ C}{1\ mole\ glucose} \right) \left(\frac{12\ g\ C}{1\ mole\ C} \right) = 110g\ C$

C09. $96\ g\ (1\ mole/12\ g) = 8\ moles\ C$ $8\ C: 8\ O: 16\ H$ in 4 moles
 $128\ g\ (1\ mole/16\ g) = 8\ moles\ O$ formula = $C_2O_2H_4 = CH_3COOH$
 $16\ g\ (1\ mole/1\ g) = 16\ moles\ H$



C13. $5\ NaN_3 + NaNO_3 \rightarrow 3\ Na_2O + 8\ N_2$ $1.50\ g\ N_2 \left(\frac{1\ mole\ N_2}{28.02\ g} \right) \left(\frac{3\ mole\ Na_2O}{8\ mole\ N_2} \right) \left(\frac{62\ g}{1\ mole\ Na_2O} \right) = 1.25\ g$

C14. $6\ Li + N_2 \rightarrow 2\ Li_3N$ 3 moles of lithium and 3 moles of nitrogen available, react in a 6 : 1 ratio
Lithium is limiting. 3 moles lithium(1 mole nitrogen/6 moles lithium) = 0.5 moles nitrogen used
3 moles – 0.5 moles = 2.5 moles nitrogen remaining

C15. KNO_3 $15.1\ g\ (1\ mole/101.11\ g) = 0.149\ moles$ $0.149\ moles/0.250\ kg\ water = 0.597\ m$

P01-P03 The God Equation, Dr. Michio Kaku, Doubleday, New York, 2021

P01. page 10, “He left a lasting legacy with his telescope, perhaps the most revolutionary and seditious invention in all of science.”

P02. page 33, in 1903, Albert Einstein secured a job as a clerk, third class in the patent office in Bern, Switzerland. This job allowed him to work on several scientific interests he had. In 1905, he published four important scientific papers. These papers explained the photoelectric effect, Brownian motion, the mass-energy equivalence and special relativity.

P03. page 66, Werner Heisenberg is credited for developing the “celebrated uncertainty principle, which stated that you cannot know precisely the velocity and location of an electron.”

P04. 21st Century Astronomy, Laura Kay, Stacy Palen, Brad Smith, and George Blumenthal; W.W. Norton; 2013, page 17, The International Astronomical Union officially redefined what astronomers mean when they call something a planet. Under this new definition, Pluto became a ‘dwarf planet along with Ceres and Eris.”

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P05. II, IV, I, III

P06. instantaneous speed

P07.

$$\sum F_x = ma, T - F_f = ma$$

$$T - \mu mg = ma$$

$$T - (\mu)(6)(9.8) = (6)(.665)$$

$$\sum F_y = Ma, Mg - T = Ma$$

$$(2.4)(9.8) - T = (2.4)(.665)$$

$$\mu = 0.305$$

$$\omega = \frac{v}{r} = \frac{4.2}{.48} = 8.75$$

$$a = \frac{v - v_0}{t} = \frac{12.6 - 4.2}{3} = 2.8$$

$$\alpha = \frac{a}{r} = \frac{2.8}{.48} = 5.8\bar{3}$$

$$\theta = \omega t + .5\alpha t^2$$

$$\beta = 10 \log \left(\frac{I}{I_0} \right)$$

$$76 = 10 \log \left(\frac{I}{10^{-12}} \right)$$

$$h = 3.36 \sin(34.5^\circ) = 1.903124956$$

$$U = K, mgh = .5mv^2 + .5I\omega^2$$

$$P08. I = .4mr^2, \omega = \frac{v}{r}, v = \sqrt{\frac{10gh}{7}}$$

$$v = \sqrt{\frac{(10)(9.8)(h)}{7}} = 5.16 \text{ m/s}$$

$$P09. \theta = (8.75)(3) + (.5)(5.8\bar{3})(3)^2$$

$$\theta = 52.5$$

$$d = \theta r = (52.5)(.48) = 25.2$$

The rock travels 25.2 m in a circular path and 25.2 m linearly as the bike travels down the road for a total of 50.4 m

$$P10. I = .000039811$$

$$\beta = 10 \log \left(\frac{12 \cdot I}{10^{-12}} \right)$$

$$\beta = 86.8 \text{ dB}$$

$$E = mc_i \Delta T_1 + mL_f + mc_w \Delta T_2 + mL_v + mc_s \Delta T_3$$

$$P11. E = (.5)(2050(15) + 335000 + 4182(100) + 2,257,000 + 2080(15))$$

$$E = 1,540,000 \text{ J}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{36} + \frac{1}{48}$$

$$C = 20.571428 \times 10^{-3}$$

P12.

$$I = \frac{\xi}{R} \left(e^{\frac{-t}{RC}} \right) = \frac{24}{120} \left(e^{\frac{-3}{(120)(C)}} \right)$$

$$I = .0593 \text{ A} \rightarrow 59.3 \text{ mA}$$

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$$E = \frac{kQ}{r^2}$$

$$E_1 = \frac{(8.99 \times 10^9)(9.22 \times 10^{-6})}{(2)^2}$$

P13. $E_1 = 20,721.95 \text{ N/C, up}$

$$E_2 = \frac{(8.99 \times 10^9)(3.36 \times 10^{-6})}{(3)^2}$$

$$E_2 = 3,356.2\bar{6} \text{ N/C, down}$$

$$E_1 + E_2 = 17,400 \text{ N/C, up}$$

$$P14. \quad F = \frac{\mu_0 I_1 I_2 L}{2\pi r} = \frac{(2 \times 10^{-7})(24)(24)(12)}{.16}$$

$$F = 0.00864 \text{ N}$$

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{24} = \frac{1}{36} + \frac{1}{d_i}, \quad d_i = 72$$

$$\xi = NBA\omega, \quad \omega = 2\pi f$$

P15. $\xi = (40)(.22)(800 \times 10^{-4})(2\pi)(60)$

$$\omega = 265 \text{ V}$$

P16. $M = -\frac{d_i}{d_o} = -\frac{72}{36} = -2$

Image is real,

inverted, enlarged

P17. $v = \frac{0.7 + 0.8}{1 + (.7)(.8)} = 0.962c$

$$(1, 15) \quad (9, 134)$$

$$\text{slope} = \frac{134 - 15}{9 - 1} = 14.875$$

P18. Alpha decay

P19. $v^2 = 2gh$

$$\text{slope} = 14.875 = 2g$$

$$g = 7.44 \text{ m/s}^2$$

$$90 - 54 = 36$$

P20. $n_1 \sin \theta_1 = n_2 \sin \theta_2$
 $(1) \sin(36^\circ) = 1.54 \sin \theta_2$

$$\theta_2 = 22.4^\circ$$